CHAPTER 11

TRENDS AND ISSUES IN ICT

CHAPTER ESSENCE AT A GLANCE:
Rapid advances in cloud computing, big data analytics, and the Internet of Things empower global digital transformation, demanding rigorous intellectual property protection, cybersecurity defense, legal accountability, and green computing stewardship.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Cloud Computing Architectures and Service Tiers:** Cloud computing delivers on-demand computing resources over the internet across Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) models under public, private, and hybrid deployment.

• **Emerging Computational Paradigms: Big Data and IoT:** Modern computing analyzes massive multi-format datasets defined by the 4 V's (Volume, Velocity, Variety, Veracity) and interconnects physical sensor-enabled objects through Internet of Things architectures.

• **Cybersecurity Threats and Intellectual Property Rights:** Protecting digital assets against malicious exploits like phishing, ransomware, and denial-of-service requires layered defense, backed by legal frameworks like the IT Act 2000 and IPR protections including patents, copyrights, and trademarks.

• **Environmental Stewardship and Green Computing:** Sustainable technology practices minimize electronic waste, promote hazardous chemical abatement (lead, cadmium, mercury), optimize hardware power efficiency, and enforce safe recycling lifecycles.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Cloud Service Delivery Stack} = \frac{\text{SaaS (Software: Google Docs, Gmail - End User)}}{\text{PaaS (Platform: Heroku, AWS Elastic Beanstalk - Developer)}} \longrightarrow \text{IaaS (Infrastructure: AWS EC2, Azure VMs)}$$

Three-Tier Cloud Service Hierarchy

$$\text{Big Data Multi-Dimensional Metrics} = \text{Volume (Petabytes)} \times \text{Velocity (Real-Time Streams)} \times \text{Variety (Structured / Unstructured)} \times \text{Veracity (Data Accuracy)}$$

Big Data 4 V's Characterization

$$\text{Cyber Threat Defense Matrix} = \frac{\text{Phishing: Deceptive Spoofing,} \quad \text{DoS: Bandwidth Flooding Exhaustion}}{\text{Malware: System Hostage Infection,} \quad \text{Identity Theft: Fraudulent Impersonation}}$$

Cybersecurity Vulnerability Classification

$$\text{Green Computing Stewardship} = \frac{\text{Energy Efficiency} \uparrow \quad \text{Toxic Heavy Metal Leaching} \downarrow}{\text{Hardware Virtualization} \uparrow \quad \text{Certified E-Waste Recycling} \uparrow}$$

Environmental Sustainability Formulations

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** SaaS provides ready-to-use software applications over the internet to end users; PaaS provides development runtimes to programmers; IaaS provides raw virtualized hardware infrastructure.

• **Alert:** Never dispose of electronic waste in municipal trash; toxic heavy metals like lead, mercury, and cadmium leach into groundwater and soil, causing severe neurological and biological toxins.

• **Alert:** The Information Technology Act 2000 (amended 2008) provides legal recognition for electronic documents and digital signatures, and prescribes criminal penalties for hacking and data theft.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between IaaS, PaaS, and SaaS cloud computing service models with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Infrastructure as a Service (IaaS): Provides raw virtualized computing infrastructure (servers, storage, networking) over the web (e.g. Amazon Web Services EC2). **[0.75 Mark]**

• Platform as a Service (PaaS): Delivers hardware and software application development platforms allowing developers to build and deploy software without managing server hardware (e.g. Google App Engine). **[0.75 Mark]**

• Software as a Service (SaaS): Delivers complete, ready-to-use end-user applications over the Internet through web browsers (e.g. Gmail, Google Docs). **[0.5 Mark]**

2. Explain the 4 V's that characterize Big Data: Volume, Velocity, Variety, and Veracity. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Volume & Velocity: Volume refers to the massive petabyte-scale size of data generated; Velocity denotes the real-time speed at which data streams are produced and processed. **[1 Mark]**

• Variety & Veracity: Variety represents diverse data formats (structured SQL, unstructured text, audio, video); Veracity measures the trustworthiness, accuracy, and quality of data. **[1 Mark]**

3. What is the Internet of Things (IoT)? Describe three practical real-world applications of IoT in smart homes and agriculture. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Definition: A network of physical objects ('things') embedded with sensors, microcontrollers, and communication hardware that collect and exchange data autonomously over the Internet. **[1 Mark]**

• Smart Homes: Automated lighting, smart thermostats, and internet-connected security locks controllable remotely from smartphones to conserve energy. **[1 Mark]**

• Smart Agriculture: Soil moisture and temperature sensors that trigger precision automated drip irrigation and monitor crop health in real time. **[1 Mark]**

4. Explain three common cyber threats: Phishing, Denial of Service (DoS) attacks, and Ransomware, detailing how they disrupt users. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Phishing: Deceptive fraudulent communications (emails, fake websites) designed to trick victims into revealing sensitive passwords, PINs, or credit card numbers. **[1 Mark]**

• Denial of Service (DoS): Flooding a web server or network with overwhelming illegitimate traffic requests to exhaust bandwidth and crash services for genuine users. **[1 Mark]**

• Ransomware: Malicious software that stealthily encrypts user files and locks the operating system, demanding extortionate financial ransom payments to restore access. **[1 Mark]**

5. Discuss Intellectual Property Rights (IPR). Distinguish between Copyright, Patent, and Trademark with suitable examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Copyright: Legal protection granted to creators of original literary, artistic, dramatic, musical, and computer software code works (e.g. software source code copyright). **[1 Mark]**

• Patent: Exclusive legal monopoly granted to an inventor for a novel technological invention or manufacturing method for a limited period (e.g. new microchip architecture). **[1 Mark]**

• Trademark: Distinctive registered name, logo, word, symbol, or brand design used to identify and distinguish products or services of an enterprise (e.g. Apple logo). **[1 Mark]**

6. What is the Information Technology Act 2000 in India? Explain two key legal provisions it provides for electronic governance and cybercrime penalties. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Significance: The primary legislation in India providing legal recognition for electronic commerce, digital documentation, and cybercrime prosecution. **[1 Mark]**

• Legal Recognition: Grants electronic records and digital signatures the same valid legal status as physical handwritten documents under Indian law. **[1 Mark]**

• Penal Provisions: Prescribes strict criminal penalties (fines and imprisonment) for hacking, identity theft, cyber terrorism, publishing obscene materials, and unauthorized data tampering. **[1 Mark]**

7. What is Green Computing? Explain practical steps for achieving green computing in educational institutions, and describe the environmental hazards caused by toxic chemicals in e-waste. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Green Computing Definition: The environmentally responsible and eco-friendly design, manufacturing, operation, and disposal of computers, servers, and associated subsystems with minimal environmental impact. **[1 Mark]**

• Practical Institutional Steps: Activating power-saving sleep modes on computer lab PCs, using energy-efficient Energy Star hardware, promoting paperless digital assessments, and virtualizing servers to reduce power draw. **[1 Mark]**

• Hazardous E-Waste Toxins: Discarded electronics contain toxic heavy metals including lead (causes brain and kidney damage), mercury (central nervous system toxin), and cadmium (lethal bio-accumulation in bones). **[1 Mark]**

• Safe Disposal Protocols: Disposing of decommissioned electronics exclusively through certified government e-waste recyclers rather than dumping in municipal landfills to prevent soil and groundwater leaching. **[1 Mark]**